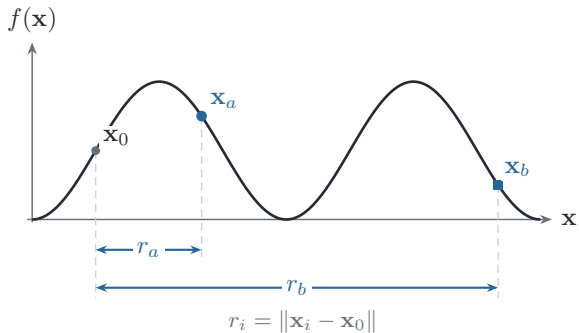


**(a) Distance-dependent noise: q-BG**

$$\mathbb{E}_{\xi} \|\zeta(\mathbf{x}; \xi)\|^p \leq Bq(\|\mathbf{x} - \mathbf{x}_0\|) + G$$

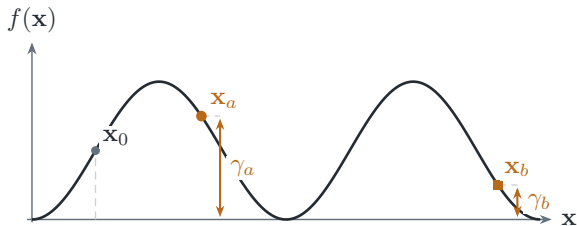


$$r_a \leq r_b \implies q(r_a) \leq q(r_b)$$

Nondecreasing  $q$  orders the permitted moment budgets.

**(b) Gap-dependent noise: q-ABC**

$$\mathbb{E}_{\xi} \|\zeta(\mathbf{x}; \xi)\|^p \leq Bq(f(\mathbf{x}) - f_*) + G$$



$$f_* := \inf_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x})$$

$$\gamma_i = f(\mathbf{x}_i) - f_*$$

$$\gamma_b \leq \gamma_a \implies q(\gamma_b) \leq q(\gamma_a)$$